EMCH 367 Controls

Dr. J. Lee

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Exam #1

Problem 1. (20pt) True or False

1. Mathematical modeling of LTI system does not vary over time.
2. Principle of superposition holds for linear system.
3. Closed loop system does not need sensors.
4. Complementary solution of spring-damper system is given as an exponential function.
5. Mass-spring-damper system is a second order system.
6. Laplace transformation is a nonlinear operator.
7. Laplace transform of a unit step function is $1/s$.
8. Laplace transform of differentiation does not need initial conditions.
9. Final value theorem, $\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s)$, is true for any $f(t)$.
10. Transfer function is the Laplace transform of the output to the Laplace transform of the input

Number	T/F	Number	T/F
1	T	6	F
2	T	7	T
3	F	8	F
4	T	9	F
5	T	10	T

Problem 2. (20pt) Find the Laplace transformation of $f(t)$ given in Figure 1.

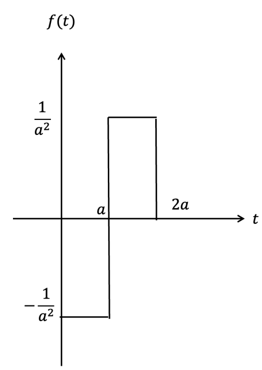

Figure 1

Solution: First, note that

$$f(t) = -\frac{1}{a^2}1(t) + \frac{2}{a^2}1(t-a) - \frac{1}{a^2}1(t-2a),$$

and hence,

$$\begin{aligned}\mathcal{L}[f(t)] &= -\frac{1}{a^2}\frac{1}{s} + \frac{2}{a^2}\frac{1}{s}e^{-as} - \frac{1}{a^2}\frac{1}{s}e^{-2as} \\ &= \frac{-1 + 2e^{-as} - e^{-2as}}{a^2 s}.\end{aligned}$$

Problem 3. (20pt) Find the inverse Laplace transformation of

$$G(s) = \frac{s}{(s+1)(s+2)^2}$$

Solution: To use partial fraction expansion,

$$G(s) = \frac{s}{(s+1)(s+2)^2} = \frac{a_1}{s+1} + \frac{a_2}{s+2} + \frac{a_3}{(s+2)^2}.$$

Now, note that

$$\begin{aligned}(s+1)G(s) &= \frac{s}{(s+2)^2} = a_1 + (s+1)R_2(s),\\ (s+2)^2 G(s) &= \frac{s}{s+1} = a_3 + (s+2)a_2 + (s+2)^2 R_4(s),\end{aligned}$$

and hence,

$$a_1 = \frac{-1}{1^2} = -1, \quad a_3 = \frac{-2}{-1} = 2.$$

Moreover, note that

$$\frac{d}{ds}\left[\frac{s}{s+1}\right] = \frac{1}{(s+1)^2} = a_2 + 2(s+2)R_4(s) + (s+2)^2 R_4'(s),$$

and hence,

$$a_2 = \frac{1}{(-1)^2} = 1.$$

Therefore, we have

$$G(s) = \frac{-1}{s+1} + \frac{1}{s+2} + \frac{2}{(s+2)^2},$$

and hence, it follows from the lookup table that

$$\mathcal{L}^{-1}[G(s)] = -e^{-t} + e^{-2t} + 2te^{-2t}.$$

Problem 4. (20pt) Consider a function $x(t)$. Show that

$$\lim_{t\to 0} \ddot{x}(0) = \lim_{s\to\infty} \left[s^3 X(s) - s^2 x(0) - s\dot{x}(0)\right]$$

Solution: Let $y(t) = \ddot{x}(t)$. Taking the Laplace transformation yields

$$Y(s) = s^2 X(s) - s x(0) - \dot{x}(0).$$

Now, it follows from initial value theorem that

$$\begin{aligned}
\lim_{t\to 0} y(t) &= \lim_{s\to\infty} sY(s) \\
&= \lim_{s\to\infty} s[s^2 X(s) - s x(0) - \dot{x}(0)] \\
&= \lim_{s\to\infty} \left[s^3 X(s) - s^2 x(0) - s\dot{x}(0)\right] \\
&= \lim_{t\to 0} \ddot{x}(t).
\end{aligned}$$

Problem 5. (20pt)

(a) (10pt) Obtain a mathematical model for the system shown in Figure 2.

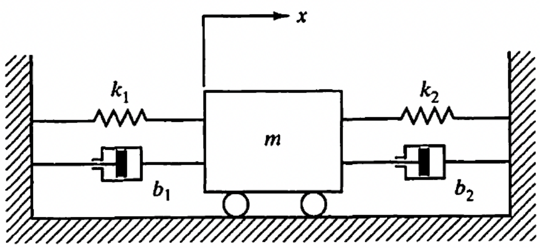

Figure 2

(b) (10pt) Let $m = 1$ kg, $b_1 = 2$ kg/s, $b_2 = 4$ kg/s, $k_1 = 10$ kg/s^2, and $k_2 = 15$ kg/s^2. Find the natural frequency and damping ratio of the system.

Solution: (a) Note that

$$m\ddot{x} = -(k_1 + k_2)x - (b_1 + b_2)\dot{x},$$

and hence,

$$m\ddot{x} + (b_1 + b_2)\dot{x} + (k_1 + k_2)x = 0.$$

(b) Now, it follows that

$$\ddot{x} + 6\dot{x} + 25x = 0,$$

and hence, Laplace transformation yields

$$[s^2 + 6s + 25]X(s) = \dot{x}(0) + 6x(0).$$

Therefore, the natural frequency is given by

$$\omega_n = 5 \text{ rad/s}.$$

Finally, the damping ratio is given by

$$\zeta = \frac{6}{2\omega_n} = 0.6.$$

Bonus Problem (20pt) Consider the system with the following transfer function.

$$G(s) = \frac{1}{2s+1}, \quad X(s) = G(s)F(s).$$

For ramp input, that is, $f(t) = t$,

(a) (10pt) find $x(t)$.

(b) (10pt) find $\lim_{t\to\infty}[x(t) - t]$.

Solution: Note that

$$F(s) = \frac{1}{s^2},$$

and hence, it follows that

$$X(s) = G(s)F(s) = \frac{1}{s^2(2s+1)}.$$

Taking inverse Laplace transform yields

$$\begin{aligned} x(t) &= \mathcal{L}^{-1}[X(s)] \\ &= \mathcal{L}^{-1}\left[\frac{1}{s^2(2s+1)}\right] \\ &= t - 2(1 - e^{-t/2}). \end{aligned}$$

Therefore, it follows that

$$\lim_{t\to\infty}[x(t) - t] = -2.$$

Look up table

	$f(t)$	$F(s)$
1	$\delta(t)$	1
2	$1(t)$	$\frac{1}{s}$
3	t	$\frac{1}{s^2}$
4	$\frac{t^{n-1}}{(n-a)!}$	$\frac{1}{s^n}$
5	t^n	$\frac{n!}{s^{n+1}}$
6	e^{-at}	$\frac{1}{s+a}$
7	te^{-at}	$\frac{1}{(s+a)^2}$
8	$\frac{1}{(n-1)!}t^{n-1}e^{-at}$	$\frac{1}{(s+a)^n}$
9	$t^n e^{-at}$	$\frac{n!}{(s+a)^{n+1}}$
10	$\sin \omega t$	$\frac{\omega}{s^2+\omega^2}$
11	$\cos \omega t$	$\frac{s}{s^2+\omega^2}$
12	$\sinh \omega t$	$\frac{\omega}{s^2-\omega^2}$
13	$\cosh \omega t$	$\frac{s}{s^2-\omega^2}$
14	$\frac{1}{a}(1-e^{-at})$	$\frac{1}{s(s+a)}$
15	$\frac{1}{b-a}(e^{-at}-e^{-bt})$	$\frac{1}{(s+a)(s+b)}$
16	$\frac{1}{b-a}(be^{-bt}-ae^{-at})$	$\frac{s}{(s+a)(s+b)}$
17	$\frac{1}{ab}\left[1+\frac{1}{a-b}(be^{-at}-ae^{-bt})\right]$	$\frac{1}{s(s+a)(s+b)}$
18	$\frac{1}{a^2}(1-e^{-at}-ate^{-at})$	$\frac{1}{s(s+a)^2}$
19	$\frac{1}{a^2}(at-1+e^{-at})$	$\frac{1}{s^2(s+a)}$
20	$e^{-at}\sin \omega t$	$\frac{\omega}{(s+a)^2+\omega^2}$

	$f(t)$	$F(s)$
21	$e^{-at}\cos\omega t$	$\frac{s+a}{(s+a)^2+\omega^2}$
22	$\frac{\omega_n}{\sqrt{1-\zeta^2}}e^{-\zeta\omega_n t}\sin\omega_n\sqrt{1-\zeta^2}t$	$\frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$
23	$-\frac{1}{\sqrt{1-\zeta^2}}e^{-\zeta\omega_n t}\sin(\omega_n\sqrt{1-\zeta^2}t-\varphi)$ $\varphi=\tan^{-1}\frac{\sqrt{1-\zeta^2}}{\zeta}$	$\frac{s}{s^2+2\zeta\omega_n s+\omega_n^2}$
24	$1-\frac{1}{\sqrt{1-\zeta^2}}e^{-\zeta\omega_n t}\sin(\omega_n\sqrt{1-\zeta^2}t+\varphi)$ $\varphi=\tan^{-1}\frac{\sqrt{1-\zeta^2}}{\zeta}$	$\frac{\omega_n^2}{s(s^2+2\zeta\omega_n s+\omega_n^2)}$
25	$1-\cos\omega t$	$\frac{\omega^2}{s(s^2+\omega^2)}$
26	$\omega t-\sin\omega t$	$\frac{\omega^2}{s^2(s^2+\omega^2)}$
27	$\sin\omega t-\omega t\cos\omega t$	$\frac{2\omega^3}{(s^2+\omega^2)^2}$
28	$\frac{1}{2\omega}t\sin\omega t$	$\frac{s}{(s^2+\omega^2)^2}$
29	$t\cos\omega t$	$\frac{s^2-\omega^2}{(s^2+\omega^2)^2}$
30	$\frac{1}{\omega_2^2-\omega_1^2}(\cos\omega_1 t-\cos\omega_2 t),\quad \omega_1\neq\omega_2$	$\frac{s}{(s^2+\omega_1^2)(s^2+\omega_2^2)}$
31	$\frac{1}{2\omega}(\sin\omega t+\omega t\cos\omega t)$	$\frac{s^2}{(s^2+\omega^2)^2}$